Template

Studentnames and studentnumbers here

2026-06-15

Set-up your environment

```r
require(tidyverse)
```

```
## Loading required package: tidyverse
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
require(rmarkdown)
```

```
## Loading required package: rmarkdown
```

```r
require(yaml)
```

```
## Loading required package: yaml
```

```r
require(WDI)
```

```
## Loading required package: WDI
```

```r
require(eurostat)
```

```
## Loading required package: eurostat
```

```r
require(ggrepel)
```

```
## Loading required package: ggrepel
```

Title Page

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Tutorial 6 Group 2

Simon Loop

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

Artificial Intelligence (AI) is changing the way how firms hire workers, manage and organise work and use technology which has led to an increasingly important social and economic issue. With recent developments in generative AI, especially after the release of Chatgpt, in Novemeber of 2022. AI tools are more visible and widely used across different industries. It is argued that AI in the workplace increase productivity but it also may reduce demand of some type of labour, especially in entry level jobs (Cazzaniga et al ., 2024).

This becomes increasingly relevant for young workers aged 18-24. As workers in this age group, often have low to nil level of work experience and are very likely to work in temporary, entry level jobs. Therefore, this results in these workers being vulnurable when employers change the skills that they demand or when certain tasks can be fulfilled by AI. Youth Employement is therefore a social problem, because unemployement in early stages of their career/life can negatively affect future earnings,skill development and work experience which is required for future higher skilled jobs (International Labour Organisation, 2024).

This project examines whether the rise of national AI-Development capcity is linked and associated with worse labour outcomes for young workers compared with the total labour force. We compare data from countries from the year 2021-2025. The aim is whether there is an observable association between AI adoption and youth unemployement relative to overall unemployement.

Cazzaniga, M., Jaumotte, F., Li, L., Melina, G., Panton, A. J., Pizzinelli, C., Rockall, E. J., Tavares, M. M., & Yao, K. (2024)

Gen-AI: Artificial intelligence and the future of work. International Monetary Fund. https://www.imf.org/en/Publications/Staff-Discussion-Notes/Issues/2024/01/14/Gen-AI-Artificial-Intelligence-and-the-Future-of-Work-542379

International Labour Organization. (2024). *Global employment trends for youth 2024.* International Labour Organization. https://www.ilo.org/publications/major-publications/global-employment-trends-youth-2024

Include the following:

- Why is this relevant?
- ...

Part 2 - Data Sourcing

2.1 Load in the data

```r
options(timeout = 600)
# Eurostat
ai_raw <- get_eurostat("isoc_eb_ai", time_format = "num")
```

```
## Dataset query already saved in cache_list.json...

## Reading cache file C:\Users\Monis\AppData\Local\Temp\RtmpI5POzr/eurostat/12bc92312dcdcb532d8c523c667

## Table  isoc_eb_ai  read from cache file:  C:\Users\Monis\AppData\Local\Temp\RtmpI5POzr/eurostat/12bc
```

```r
# Filter out EU countries, fix the 2 misaligned EU codes and drop aggregates
eu_codes <- ai_raw %>%
  distinct(geo) %>%
  mutate(iso2c = recode(geo, "EL" = "GR", "UK" = "GB")) %>%
  filter(nchar(iso2c) == 2, iso2c != "EA") %>%
  pull(iso2c)

# World Bank: download ONCE, then reuse the saved copy
if (file.exists("wb_raw.rds")) {
  wb_raw <- readRDS("wb_raw.rds")
} else {
  wb_raw <- WDI(
    country   = eu_codes,
    indicator = c(total_unemp = "SL.UEM.TOTL.ZS",
                  youth_unemp = "SL.UEM.1524.ZS"),
    start = 1991, end = 2024, extra = TRUE
  )
  saveRDS(wb_raw, "wb_raw.rds")
}
```

2.2 Provide a short summary of the dataset(s)

```r
head(ai_raw)
```

```
## # A tibble: 6 x 8
##   freq  size_emp nace_r2      indic_is        unit      geo   TIME_PERIOD values
##   <chr> <chr>    <chr>        <chr>           <chr>     <chr>       <dbl>  <dbl>
## 1 A     0-9      C10-S951_X_K E_AIX_CC1SIX_DA PC_ENT    ES           2023   5.74
## 2 A     0-9      C10-S951_X_K E_AIX_CC1SIX_DA PC_ENT    ES           2024   5.66
## 3 A     0-9      C10-S951_X_K E_AIX_CC1SIX_DA PC_ENT    ES           2025   5.49
## 4 A     0-9      C10-S951_X_K E_AIX_CC1SIX_DA PC_ENT    PT           2023  13.4
## 5 A     0-9      C10-S951_X_K E_AIX_CC1SIX_DA PC_ENT_I~ ES           2023   6.88
## 6 A     0-9      C10-S951_X_K E_AIX_CC1SIX_DA PC_ENT_I~ ES           2024   6.94
```

```r
str(ai_raw)
```

```
## tibble [72,527 x 8] (S3: tbl_df/tbl/data.frame)
##  $ freq       : chr [1:72527] "A" "A" "A" "A" ...
```

```
##  $ size_emp   : chr [1:72527] "0-9" "0-9" "0-9" "0-9" ...
##  $ nace_r2    : chr [1:72527] "C10-S951_X_K" "C10-S951_X_K" "C10-S951_X_K" "C10-S951_X_K" ...
##  $ indic_is   : chr [1:72527] "E_AIX_CC1SIX_DA" "E_AIX_CC1SIX_DA" "E_AIX_CC1SIX_DA" "E_AIX_CC1SIX_DA
##  $ unit       : chr [1:72527] "PC_ENT" "PC_ENT" "PC_ENT" "PC_ENT" ...
##  $ geo        : chr [1:72527] "ES" "ES" "ES" "PT" ...
##  $ TIME_PERIOD: num [1:72527] 2023 2024 2025 2023 2023 ...
##  $ values     : num [1:72527] 5.74 5.66 5.49 13.36 6.88 ...
```

```
summary(ai_raw)
```

```
##        freq            size_emp          nace_r2           indic_is      
##  Length    :72527   Length    :72527   Length    :72527   Length    :72527  
##  N.unique :     1   N.unique :     8   N.unique :     1   N.unique :    63  
##  N.blank  :     0   N.blank  :     0   N.blank  :     0   N.blank  :     0  
##  Min.nchar:     1   Min.nchar:     3   Min.nchar:    12   Min.nchar:     7  
##  Max.nchar:     1   Max.nchar:     6   Max.nchar:    12   Max.nchar:    18  
##                                                                          
##                                                                          
##        unit              geo          TIME_PERIOD       values      
##  Length    :72527   Length    :72527   Min.   :2021   Min.   :  0.00  
##  N.unique :     6   N.unique :    36   1st Qu.:2023   1st Qu.:  2.31  
##  N.blank  :     0   N.blank  :     0   Median :2024   Median :  6.01  
##  Min.nchar:     6   Min.nchar:     2   Mean   :2023   Mean   : 15.43  
##  Max.nchar:    14   Max.nchar:     9   3rd Qu.:2025   3rd Qu.: 19.99  
##                                        Max.   :2025   Max.   :100.00  
##                                                       NAs    :1771
```

```
names(ai_raw)
```

```
## [1] "freq"        "size_emp"    "nace_r2"     "indic_is"    "unit"       
## [6] "geo"         "TIME_PERIOD" "values"
```

```
head(wb_raw)
```

```
##   country iso2c iso3c year status lastupdated total_unemp youth_unemp
## 1 Albania    AL   ALB 1991          2026-04-08      10.304      15.687
## 2 Albania    AL   ALB 1992          2026-04-08      30.007      45.621
## 3 Albania    AL   ALB 1993          2026-04-08      25.251      38.820
## 4 Albania    AL   ALB 1994          2026-04-08      20.835      32.402
## 5 Albania    AL   ALB 1995          2026-04-08      14.607      23.005
## 6 Albania    AL   ALB 1996          2026-04-08      13.928      22.071
##                  region capital longitude latitude              income lending
## 1 Europe & Central Asia  Tirane   19.8172  41.3317 Upper middle income    IBRD
## 2 Europe & Central Asia  Tirane   19.8172  41.3317 Upper middle income    IBRD
## 3 Europe & Central Asia  Tirane   19.8172  41.3317 Upper middle income    IBRD
## 4 Europe & Central Asia  Tirane   19.8172  41.3317 Upper middle income    IBRD
## 5 Europe & Central Asia  Tirane   19.8172  41.3317 Upper middle income    IBRD
## 6 Europe & Central Asia  Tirane   19.8172  41.3317 Upper middle income    IBRD
```

```
str(wb_raw)
```

```
## 'data.frame':    1156 obs. of  14 variables:
##  $ country    : chr  "Albania" "Albania" "Albania" "Albania" ...
##  $ iso2c      : chr  "AL" "AL" "AL" "AL" ...
##  $ iso3c      : chr  "ALB" "ALB" "ALB" "ALB" ...
##  $ year       : int  1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 ...
##  $ status     : chr  "" "" "" "" ...
##  $ lastupdated: chr  "2026-04-08" "2026-04-08" "2026-04-08" "2026-04-08" ...
##  $ total_unemp: num  10.3 30 25.3 20.8 14.6 ...
##   ..- attr(*, "label")= chr "Unemployment, total (% of total labor force) (modeled ILO estimate)"
##  $ youth_unemp: num  15.7 45.6 38.8 32.4 23 ...
##   ..- attr(*, "label")= chr "Unemployment, youth total (% of total labor force ages 15-24) (modeled I
##  $ region     : chr  "Europe & Central Asia" "Europe & Central Asia" "Europe & Central Asia" "Europe
##  $ capital    : chr  "Tirane" "Tirane" "Tirane" "Tirane" ...
##  $ longitude  : chr  "19.8172" "19.8172" "19.8172" "19.8172" ...
##  $ latitude   : chr  "41.3317" "41.3317" "41.3317" "41.3317" ...
##  $ income     : chr  "Upper middle income" "Upper middle income" "Upper middle income" "Upper middle
##  $ lending    : chr  "IBRD" "IBRD" "IBRD" "IBRD" ...
```

```r
summary(wb_raw)
```

```
##      country              iso2c               iso3c                year     
##  Length   :1156   Length   :1156   Length   :1156   Min.   :1991  
##  N.unique :  34   N.unique :  34   N.unique :  34   1st Qu.:1999  
##  N.blank  :   0   N.blank  :   0   N.blank  :   0   Median :2008  
##  Min.nchar:   5   Min.nchar:   2   Min.nchar:   3   Mean   :2008  
##  Max.nchar:  22   Max.nchar:   2   Max.nchar:   3   3rd Qu.:2016  
##                                                     Max.   :2024  
##       status          lastupdated      total_unemp      youth_unemp    
##  Length   :1156   Length   :1156   Min.   : 1.100   Min.   : 1.956  
##  N.unique :   1   N.unique :   1   1st Qu.: 5.775   1st Qu.:12.407  
##  N.blank  :1156   N.blank  :   0   Median : 8.264   Median :19.952  
##  Min.nchar:   0   Min.nchar:  10   Mean   :10.260   Mean   :22.377  
##  Max.nchar:   0   Max.nchar:  10   3rd Qu.:12.800   3rd Qu.:28.542  
##                                    Max.   :38.800   Max.   :68.882  
##       region            capital            longitude          latitude     
##  Length   :1156   Length   :1156   Length   :1156   Length   :1156  
##  N.unique :   2   N.unique :  34   N.unique :  34   N.unique :  34  
##  N.blank  :   0   N.blank  :   0   N.blank  :   0   N.blank  :   0  
##  Min.nchar:  21   Min.nchar:   4   Min.nchar:   5   Min.nchar:   5  
##  Max.nchar:  49   Max.nchar:  10   Max.nchar:   8   Max.nchar:   7  
##                                                                     
##       income            lending         
##  Length   :1156   Length   :1156  
##  N.unique :   2   N.unique :   2  
##  N.blank  :   0   N.blank  :   0  
##  Min.nchar:  11   Min.nchar:   4  
##  Max.nchar:  19   Max.nchar:  14  
## 
```

```r
names(wb_raw)
```

```
##  [1] "country"     "iso2c"       "iso3c"       "year"        "status"     
##  [6] "lastupdated" "total_unemp" "youth_unemp" "region"      "capital"    
## [11] "longitude"   "latitude"    "income"      "lending"
```

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

The AI data comes from Eurostats that shows business wealth and resources . The unemployment data comes from World Bankdata that shows the financial trouble and living standards .the two main variables are Total unemployment rate and youth unemploymentent. the youth unemployment is used as the sub-population to see how young workers are affected by AI data usage in businesses. Where: European countries only. When: Years 2021 to 2025 (but 2022 not included). This timeline captures the exact period where the post-pandemic labor market met the massive generative AI boom.

Part 3 - Quantifying

3.1 Data cleaning

Say we want to include only larger distances (above 2) in our dataset, we can filter for this.

```
wb_clean <- wb_raw %>%
  filter(year == 2021 | year == 2022 | year == 2023 | year == 2024)

ai_clean <- ai_raw %>%
  filter(indic_is == "E_AI_TANY")
```

Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

Variable 1

```
wb_clean <- wb_clean %>%
  mutate(youth_ratio = youth_unemp / total_unemp)
```

Variable 2

```
combined <- wb_clean %>%
  left_join(ai_clean, by = c("iso2c" = "geo", "year" = "TIME_PERIOD")) %>%
mutate(ai_group = case_when(
  values < 10 ~ "Low",
  values < 30 ~ "Medium",
  TRUE ~ "High"
))
names(wb_clean)
```

```
##  [1] "country"     "iso2c"       "iso3c"       "year"        "status"
##  [6] "lastupdated" "total_unemp" "youth_unemp" "region"      "capital"
## [11] "longitude"   "latitude"    "income"      "lending"     "youth_ratio"
```

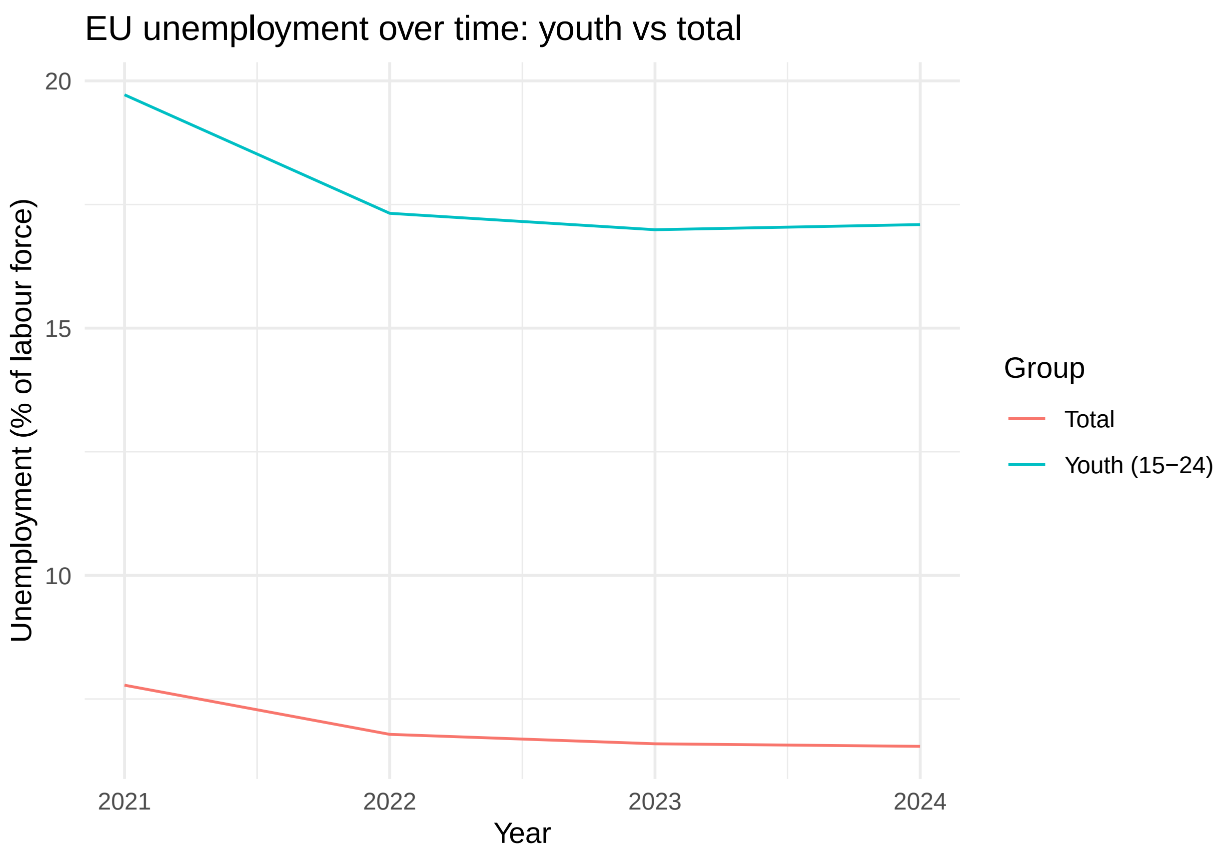
Table with new variables

```r
combined %>%
  group_by(ai_group) %>%
  summarise(
    n = n(),
    mean_total_unemp = mean(total_unemp, na.rm = TRUE),
    mean_youth_unemp = mean(youth_unemp, na.rm = TRUE),
    mean_youth_ratio = mean(youth_ratio, na.rm = TRUE)
  )
```

```
## # A tibble: 3 x 5
##   ai_group     n mean_total_unemp mean_youth_unemp mean_youth_ratio
##   <chr>    <int>            <dbl>            <dbl>            <dbl>
## 1 High       169             7.12             17.6             2.57
## 2 Low        500             7.29             19.1             2.76
## 3 Medium     373             5.76             15.3             2.70
```

3.3 Visualize temporal variation

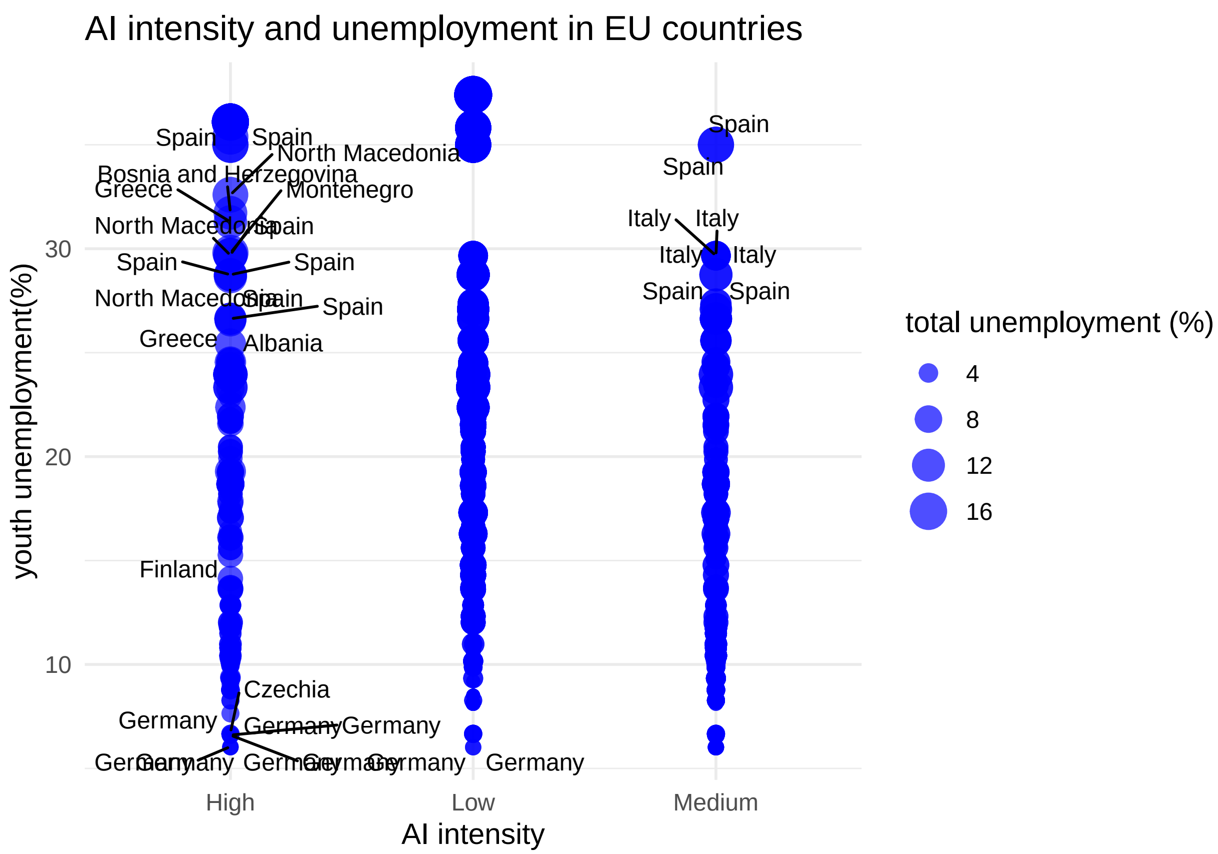
```r
# Average unemployment rate per year
yearly <- wb_clean %>%
  group_by(year) %>%
  summarise(
    mean_total_unemp = mean(total_unemp, na.rm = TRUE),
    mean_youth_unemp = mean(youth_unemp, na.rm = TRUE)
  )

# Plotting the average unemployment rate per year
ggplot(yearly, aes(x = year)) +
  geom_line(aes(y = mean_total_unemp, color = "Total")) +
  geom_line(aes(y = mean_youth_unemp, color = "Youth (15-24)")) +
  labs(title = "EU unemployment over time: youth vs total",
       x = "Year", y = "Unemployment (% of labour force)",
       color = "Group") +
  theme_minimal()
```



ai_group - bins the continuous values into Low, Medium, and High, so we can compare youth unemployment outcomes across different levels of AI adotion.

3.4 Visualize spatial variation

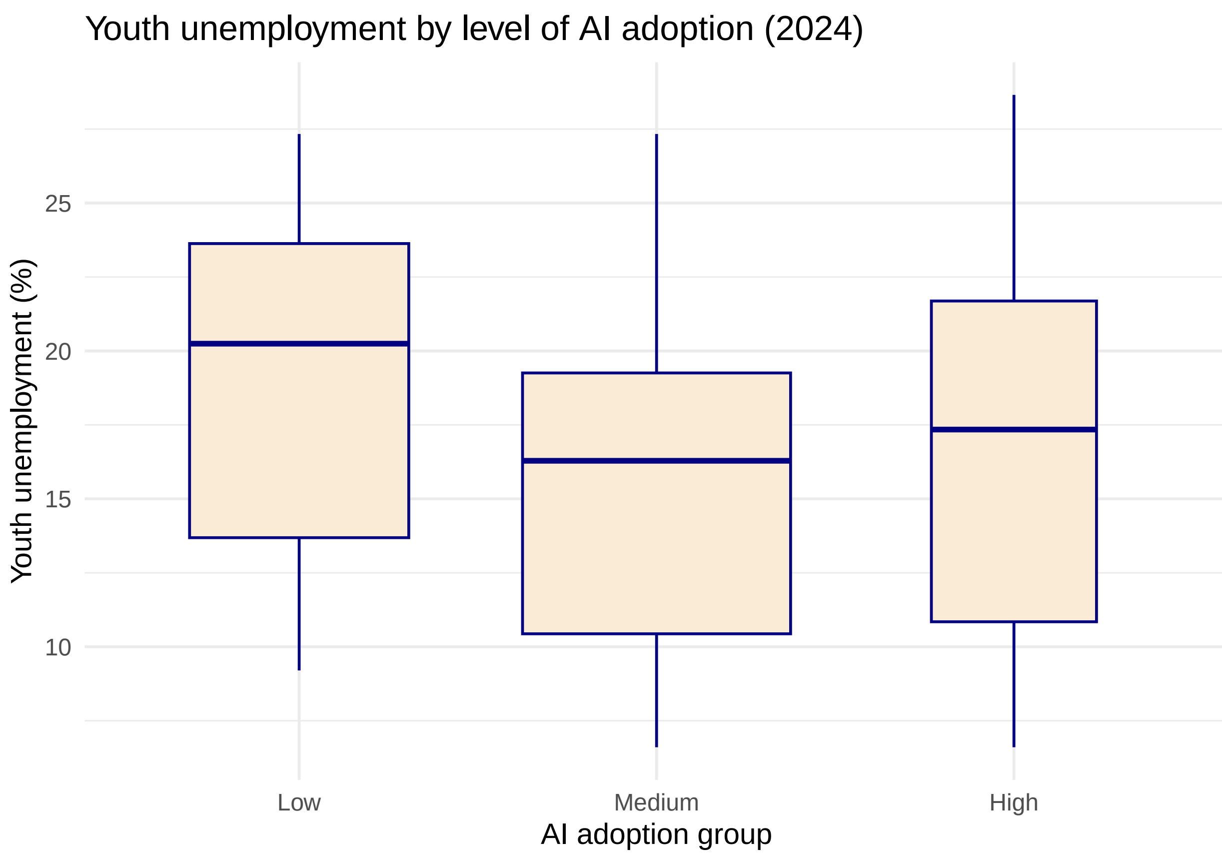
```
ai_group <- combined %>%
   filter(year == 2024, !is.na(ai_group)) %>%
  mutate(ai_group = factor(ai_group, levels = c("Low", "Medium", "High")))

ggplot (combined, aes(x=ai_group,
                    y= youth_unemp,
                    size= total_unemp,
                    label=country)) +
  geom_point(alpha=0.7, color = "blue") +
  geom_text_repel(size=3, max.overlaps= 20)+
  scale_size_continuous(name="total unemployment (%)") +

  labs(x="AI intensity",
       y="youth unemployment(%)",
       title= "AI intensity and unemployment in EU countries") + theme_minimal()
```



3.5 Visualize sub-population variation

What is the poverty rate by state?

```
combined %>%
  filter(year == 2024, !is.na(ai_group)) %>%
  mutate(ai_group = factor(ai_group, levels = c("Low", "Medium", "High"))) %>%
  ggplot(aes(x = ai_group, y = youth_unemp)) +
  geom_boxplot(na.rm = TRUE, fill = "antiquewhite", color = "navy", outlier.color = "aquamarine", varwi
  labs(title = "Youth unemployment by level of AI adoption (2024)",
       x = "AI adoption group", y = "Youth unemployment (%)") +
  theme_minimal()
```



Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

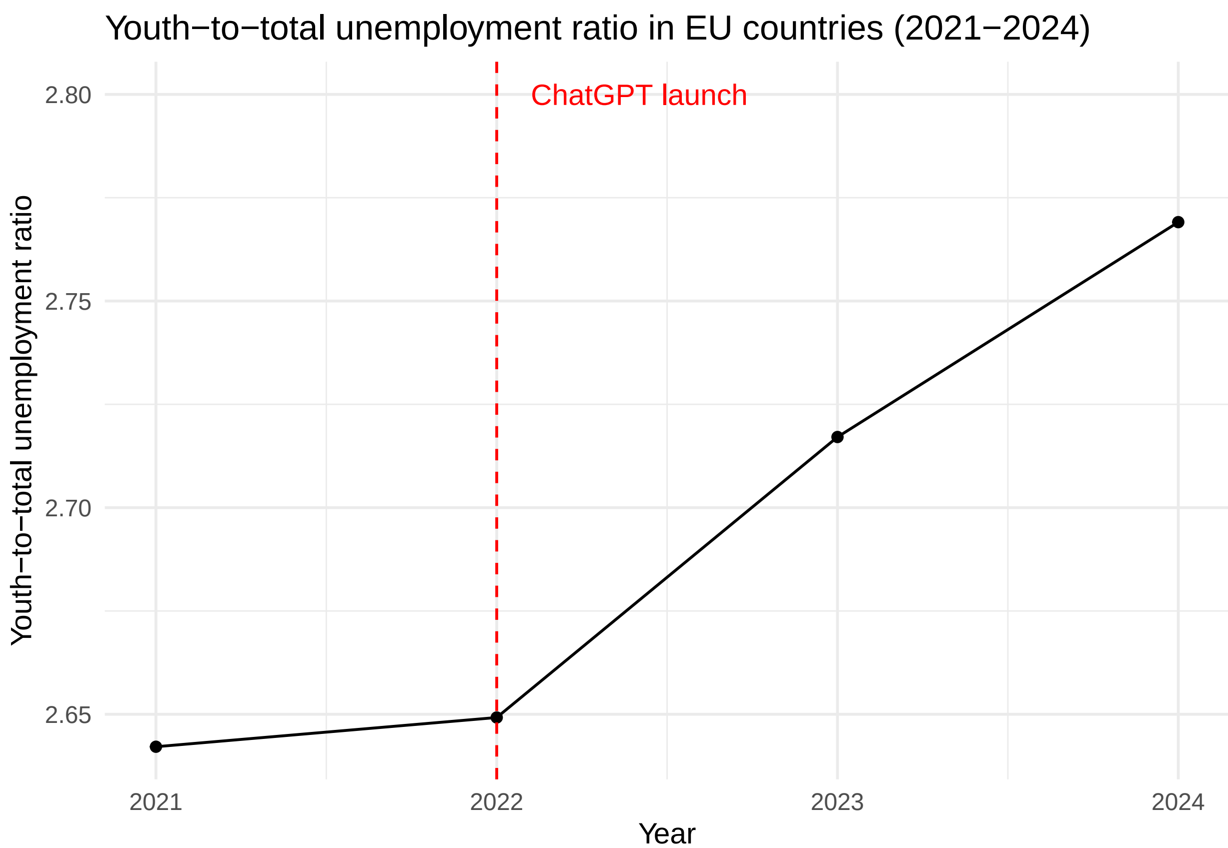
Analyze the relationship between two variables.

```
ratio_by_year <- combined %>%
  group_by(year) %>%
  summarize(
    mean_youth_ratio = mean(youth_ratio, na.rm = TRUE)
  )
ratio_by_year
```

```
## # A tibble: 4 x 2
##    year mean_youth_ratio
##   <dbl>            <dbl>
## 1  2021             2.64
## 2  2022             2.65
## 3  2023             2.72
## 4  2024             2.77
```

```
#Plotting the youth-to-total unemployment ratio per year
ggplot(ratio_by_year, aes (x = year, y = mean_youth_ratio)) +
  geom_line() +
  geom_point() +
```

```r
  geom_vline(xintercept = 2022, linetype = "dashed", color = "red") +
  annotate("text", x = 2022.1, y = 2.8, label = "ChatGPT launch", hjust = 0, color = "red") +
    labs(title = "Youth-to-total unemployment ratio in EU countries (2021-2024)",
         x = "Year",
         y = "Youth-to-total unemployment ratio") +
    theme_minimal()
```



The graph shows the average youth-to-total unemployment ratio across EU countries from 2021 to 2024. The ratio rises steadily after 2022 - the year ChatGPT launched - suggesting that young workers may be increasingly disadvantaged relative to the broader workforce as AI adoption grows. However, this is an association, not proof: other factors such as post-COVID labor market shifts or broader economic conditions could equally explain the trend.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: …

5.2 Reference list

Use APA referencing throughout your document.